Business Requirements Document

Finance Analytics & Transaction Risk Monitoring

Portfolio case study | Version 1.0 | November 2026

Document status: Proposed analytical requirements, mapped to completed project deliverables. Synthetic educational data; no real organizational sponsor or deployment is claimed.

1. Executive Context

A hypothetical financial services analytics team requires a consistent view of transaction activity, operational reliability, customer engagement and the potential workload created by transaction-risk reviews. The solution combines SQL Server validation, Python statistical analysis and Power BI reporting to support exploratory decision-making.

2. Business Problem & Objectives

ID	Business objective	Decision supported
BO-01	Measure transaction volume, value, fees and tax consistently.	Financial operations reporting
BO-02	Understand payment-channel and transaction-type failure patterns.	Reliability monitoring and investigation prioritization
BO-03	Monitor monthly failure behavior and screen for unusual elevations.	Operational surveillance
BO-04	Classify customers by recorded transaction activity.	Engagement and service-capacity planning
BO-05	Compare risk-score review thresholds and existing fraud-flag coverage.	Risk review capacity planning

3. Stakeholders & Intended Users

Stakeholder (illustrative)	Information needs
Finance / business analyst	Financial KPIs, trend and category breakdowns
Payments operations	Success/failure counts, rates, channel/type combinations
Customer analytics	Customer activity cohorts and existing segment comparisons
Risk operations	Review volume and existing flag coverage by

	threshold
Data / BI analyst	Reproducible data preparation, checks and consistent metric definitions

4. Scope & Deliverables

In scope

Educational customer and transaction CSV files; raw SQL Server tables; deduplicated and cleaned SQL views; data-quality validation; Python statistical testing, customer segmentation and risk sensitivity; Power BI report with Overview Analysis and Transactions pages; written recommendations and documentation.

Out of scope

Live transaction processing, real-time alerting, model training, production fraud decisions, customer contact campaigns, causal attribution, deployment to a financial institution and measured business savings.

5. Data Sources & Architecture

Layer	Source / artifact	Purpose
Input	data/customers.csv; data/finance_transactions.csv	Synthetic educational source records
SQL staging	raw.customers; raw.finance_transactions	Import source data without destructive transformation
SQL validation	validated.finance_transactions; validated.clean_transactions; validated.clean_customers	Deduplication, cleaning and typed analytical views
Analysis	Python Jupyter notebook	Statistical tests, cohorts and risk-threshold scenarios
Reporting	Power BI PBIX: Overview Analysis; Transactions	Interactive financial KPI reporting and transaction exploration

Observed project coverage: 5,000 customers and 50,000 cleaned transactions, January 2023–April 2026. Raw transactions contain 50,069 records; the validated transaction view returns 50,000. The SQL environment is FinanceAnalyticsDB.

6. Functional Requirements

ID	Requirement	Deliverable / verification
FR-01	Load the provided CSVs into SQL Server raw staging tables.	SQL setup, DDL and import scripts
FR-02	Remove transaction-ID	validated views and SQL

	duplicates deterministically and standardize analytical fields.	cleaning scripts
FR-03	Validate unique customer and transaction IDs and customer referential integrity.	09_data_quality_validation.sql
FR-04	Display financial totals, average transaction value, fees and tax with interactive slicing.	Power BI Overview Analysis
FR-05	Provide transaction-level drill-down and filters.	Power BI Transactions page
FR-06	Compare transaction outcomes across channels with statistical testing.	Python chi-square and effect-size analysis
FR-07	Evaluate channel/type failures and adjust exploratory significance tests for multiple comparisons.	Python failure screening
FR-08	Screen monthly elevated failure rates using a historical baseline.	Python time-series monitoring
FR-09	Segment customers by recorded transaction frequency.	Python behavioral cohort analysis
FR-10	Compare risk-score thresholds by review workload and coverage of existing flags.	Python risk-threshold sensitivity analysis

7. KPI & Metric Definitions

Metric	Definition / interpretation
Total transactions	Count of cleaned, unique transaction records
Total transaction amount	Sum of recorded transaction amount; not company revenue
Average transaction value	Total transaction amount / transaction count
Total fees / total tax	Sums of recorded fee_amount / tax_amount fields
Failure rate	Failed transactions / all cleaned transactions × 100
Channel success rate	Successful transactions / all transactions in the channel × 100
Customer activity cohort	No recorded activity, low, medium or high; active quartiles Q1=7, Q3=17

Review rate	Transactions at/above selected risk threshold / all transactions × 100
Existing flag coverage	Existing fraud-flagged transactions at/above threshold / all existing flagged transactions × 100

8. Reporting & Analytical Requirements

Power BI

The Overview Analysis page includes headline financial KPIs and breakdowns by month, transaction status, customer segment, state, gender and transaction type. The Transactions page supports transaction-level inspection. Visible filters include year, dynamic metric, occupation and merchant category. The report is primarily a financial activity dashboard, not a standalone risk-monitoring dashboard.

Python

Analyze success/failure performance by payment channel and transaction type; apply chi-square and multiple-testing procedures; screen monthly failure patterns; build behavioral cohorts; compare risk thresholds from 50 to 90. Clearly separate descriptive results, statistically supported conclusions and proposed actions.

9. Nonfunctional Requirements & Controls

ID	Requirement
NFR-01	SQL views and notebook must preserve original raw CSV files.
NFR-02	Document reproducible execution order, local SQL Server and ODBC prerequisites.
NFR-03	Check identifier uniqueness, referential integrity and expected record counts.
NFR-04	Avoid exposing customer-level personal information in published notebook outputs or screenshots.
NFR-05	Keep secrets, local credentials and environment files out of Git.
NFR-06	State that educational data and existing fraud flags do not validate real-world fraud detection.

10. Acceptance Criteria & Evidence

Criterion	Acceptance evidence / status
AC-01	50,069 raw transactions, 50,000 cleaned transactions: verified in SQL checks.
AC-02	5,000 customers, 5,000 distinct customer IDs,

	zero duplicates: verified.
AC-03	50,000 distinct cleaned transaction IDs, zero duplicates: verified.
AC-04	Zero cleaned transactions with unmatched customer IDs: verified.
AC-05	Notebook contains channel, failure, monthly, customer and risk analyses: confirmed by file inspection.
AC-06	PBIX contains Overview Analysis and Transactions pages: confirmed by file inspection.
AC-07	Fresh-environment end-to-end rebuild: not yet demonstrated; remains a reproducibility check.

11. Baseline Findings for Validation

These are observed analysis results, not targets or achieved operational improvements.

Area	Observed finding
Reliability	5,096 failed of 50,000 transactions; failure rate 10.192%.
Channel association	Chi-square 10.6981; p=0.0982; Cramér's V=0.0146; not significant at 5%.
Exploratory screens	No channel/type elevated failure rates after FDR adjustment; no significant elevated monthly rates among 34 screened months.
Engagement	896 high-activity customers (17.92%) generated 42.91% of transactions.
Risk review ≥65	4,884 transactions (9.77%); 100% of existing fraud flags covered.
Risk review ≥70	1,212 transactions (2.42%); 84.92% of existing flags covered.
Risk review ≥75	460 transactions (0.92%); 73.02% of existing flags covered.

12. Risks, Assumptions & Limitations

The case uses synthetic educational data, so results are not evidence of real-world institutional performance. Fraud flags may be dependent on risk scores and are not independently confirmed outcomes. Monthly aggregation can hide short incidents; exploratory significance testing does not establish causality. Transaction value is not revenue. Customer activity is not profitability or churn. A review threshold cannot be declared optimal without verified outcomes, staffing costs and error costs. No business intervention or financial benefit has been measured.

13. Proposed Business Use & Success Measures

If implemented in a real organization, stakeholders could track failure-rate trends, investigation queue size, review turnaround time, validated fraud outcomes, customer engagement and dashboard adoption. These are proposed future measures, not project outcomes. A pilot and independent validation would be required before policy or threshold changes.

14. Traceability & Source Artifacts

Requirements were mapped to the uploaded Finance_Analytics_Python_Analysis.ipynb and Finance_Analytics_Dashboard.pbix, together with the previously confirmed SQL data-quality checks. The BRD describes a portfolio case-study scenario rather than a formally approved stakeholder specification.